Do Now · retrieval

Lesson 104 · Magnetic, Optical and Flash Storage

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What does non-volatile mean?

Q2

What is seek time?

Q3

What does optical storage use to read data?

Q4

1 byte contains how many bits?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Understand — Magnetic hard disks use magnetised regions on spinning platters with moving heads; they can offer large capacity at relatively low cost per GB.

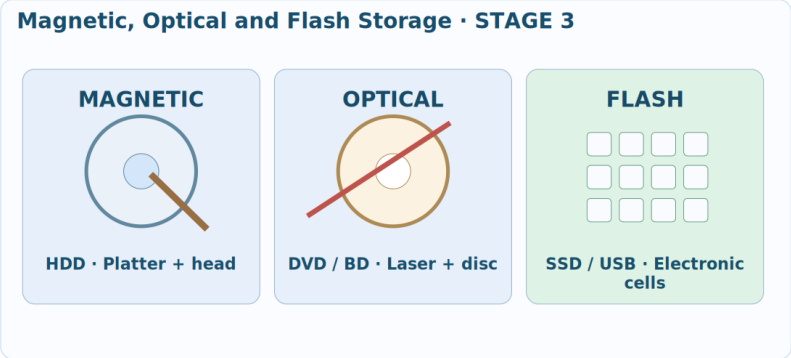
Core idea 2

Apply — Optical discs are read with a laser, often used for physically distributed or archived/removable media, though capacity and drive availability vary.

Core idea 3

Evaluate — Flash uses electronic non-volatile memory cells without moving read/write parts; often faster random access and more robust to vibration than HDDs, but trade-offs remain.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

magnetic — Storage via magnetisation.

optical — Storage read with laser/light.

flash — Non-volatile electronic solid-state storage.

cost per GB — Price relative to capacity.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

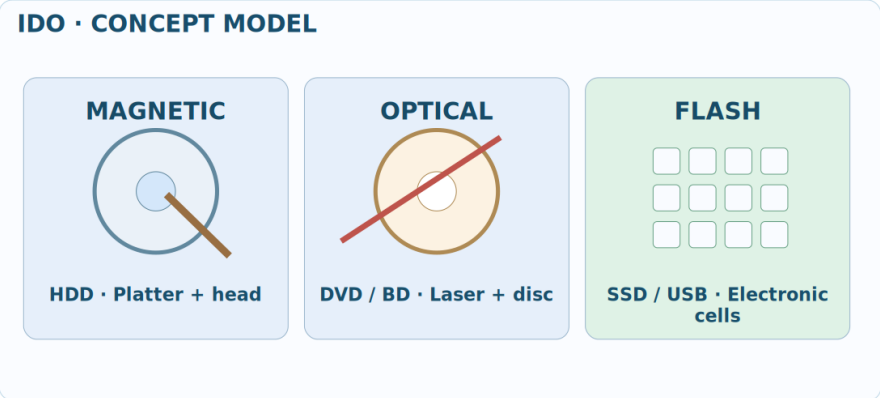
Problem

Choose portable storage for a field photographer travelling over rough ground with active projects.

Follow the worked method

1. Explain the choice — Flash SSD: no spinning platter and generally quick random access; still requires backups.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Explain the choice
Magnetic HDD is a reasonable candidate because of capacity and cost per GB; validate resilience and transfer needs.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Match HDD, DVD and USB SSD to magnetic, optical and flash storage, with mechanisms. [6]

Question 2

Explain one advantage and one limitation of flash SSD over magnetic HDD for a mobile video editor. [6]

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Evaluate storage choice for distributing 100 offline revision packs to pupils and for a multi-terabyte editable school video archive. [8]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A hard disk drive reads bits using a laser.

Use key vocabulary from memory

1. Explain magnetic in your own words.

2. Explain optical in your own words.

3. Explain flash in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.